

Circles and circle theorems - Set 2

TMUA practice questions by topic. Specification MM3.2, MM3.3, M5.8, M5.9. 5 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Find the complete set of values of k for which $x^2 + y^2 + 2kx - 4y + 3k + 4 = 0$ represents a circle.
- A** $k > 3$
B $0 < k < 3$
C $k < -3$ or $k > 0$
D $k < 0$ or $k > 3$
E $-3 < k < 0$
F all real k
- 2 The line $y = x + c$ is a tangent to the circle $(x - 1)^2 + (y - 2)^2 = 8$. Find the possible values of c .
- A** $c = -3$ or $c = 5$
B $c = 1 \pm 2\sqrt{2}$
C $c = 1$ or $c = 5$
D $c = -1$ or $c = 3$
E $c = \pm 4$
F $c = -5$ or $c = 3$
- 3 Find the length of the tangent from the point $(7, 1)$ to the circle $x^2 + y^2 - 2x - 4y - 4 = 0$.
- A** 3
B $\sqrt{28}$
C 6
D $\sqrt{26}$
E $\sqrt{37}$
F 5

- 4 Two circles have equations $x^2 + y^2 = 25$ and $(x - 8)^2 + y^2 = 9$. Which of the following statements is/are true?
- I The circles touch externally.
 - II The circles intersect at exactly two points.
 - III The point $(5, 0)$ lies on both circles.
- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III
- 5 The point $P(x, y)$ moves so that its distance from $(3, 0)$ is twice its distance from $(0, 0)$. The locus of P is a circle. Find its radius.
- A $\sqrt{2}$
- B 3
- C 2
- D 4
- E $\sqrt{3}$
- F 1